VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-01 PHYSICS

Shailendra sir



Topics To Be Covered

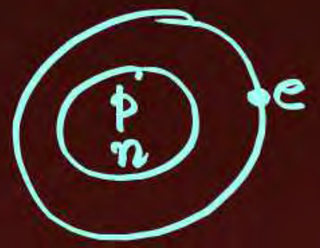
1. REVISION
2. CONCEPT OF CURRENT
3. OHM'S LAW
4. QUESTIONS


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Physics Wallah

Revision

charge
⇓
Electrical & Magnetic Property.

- Positive ⇓ Deficiency of electrons
- Negative ⇓ Excess of electrons.



Proton	⊕	$1.6\times10^{-19}C$	Rutherford/Goldstein
Neutron		$0\,C$	Chadwick
Electron	⊖	$-1.6\times10^{-19}C$	Thomson

(SSS) ईंट पर नाच



CONCEPT OF CHARGE

(SI) Ampere - second or Coulomb

(CGS) Stat - Coulomb

Scalar Quantity

Quantisation of Charge

$q = ne$

↓

integer

$n = 0, 1, 2, 3, 4 \ldots\ldots$

$n \neq$ fraction, decimal

$\left(\frac{3}{7}, 1.37\right)$

No. of electrons for 1C charge

$$q = ne$$

$$1 = n(1.6 \times 10^{-19})$$

$$n = \frac{1}{1.6 \times 10^{-19}}$$

$$= \frac{10^{19}}{1.6}$$

$$= \frac{10}{1.6} \times 10^{18}$$

$$\boxed{n = 6.25 \times 10^{18}}$$

CONCEPT OF CURRENT

Rate of flow of charge.

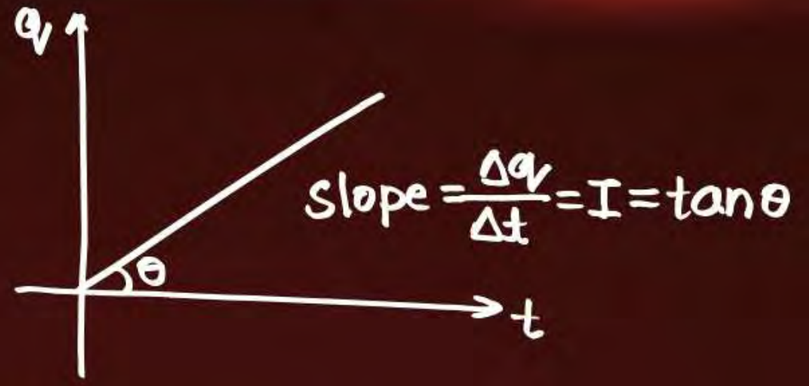
$$I = \frac{\Delta q}{\Delta t}$$

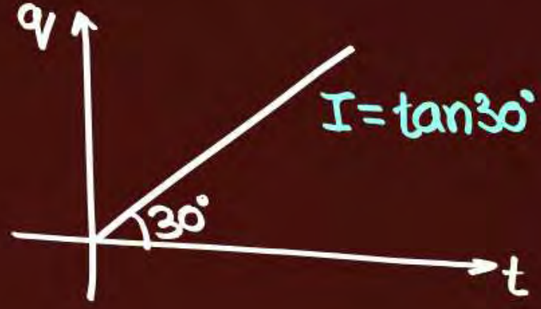


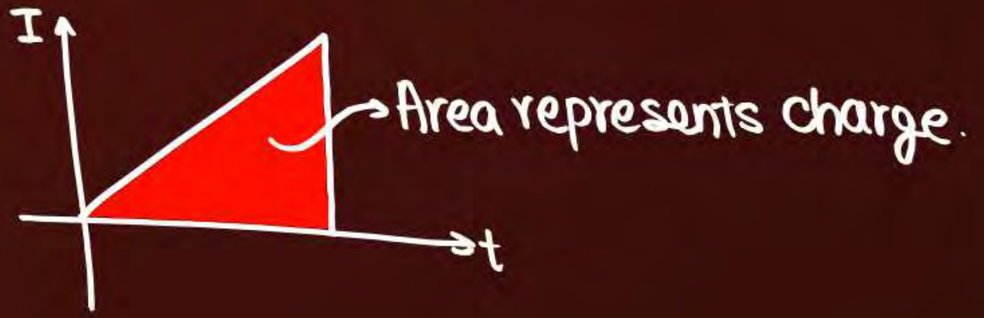
- Scalar Quantity
- C/s or Ampere

In opposite direction of electron.









CONCEPT OF POTENTIAL

⇒ Work Done per unit charge



$$V_P = \frac{W_{\infty \to P}}{q_0}$$

→ J/C or Volt

→ Scalar Quantity

→ May be +ve, -ve or zero.

$W = 100\ J$

$q_0 = 4C$



$$V = \frac{W}{q_0} = \frac{100}{4} = 25 \text{ Volt.}$$

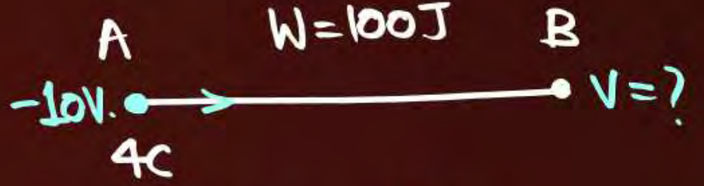
CONCEPT OF POTENTIAL DIFFERENCE

⇒ Work Done per unit charge in bringing it from one point of electric field to another point of electric field.



$$\Delta V = \frac{W_{A\to B}}{q_0}$$

$$\boxed{V_B - V_A = \frac{W_{AB}}{q_0}}$$



$$V_B - (-10) = \frac{100}{4}$$

$$V_B + 10 = 25$$

$$\boxed{V_B = 15V.}$$

CONCEPT OF RESISTANCE (Obstacles)

The obstruction offered to the flow of current by the conductor (or wire) is called its resistance.

→ Collision among electrons and atom.

$R \propto l$

$R \propto \frac{1}{A}$



On Combining,

$R \propto \frac{l}{A}$

$R = \rho \frac{l}{A}$

Specific Resistance or Resistivity

SSS

ρ → Property of Material of Conductor & temperature.

R → Depends on

(i) Material
(ii) Temperature
(iii) length
(iv) Area

Unit → ohm or Ω

$$R = \rho \frac{l}{A}$$



CONDUCTANCE

$$G = \frac{1}{R}$$

→ ohm^{-1} or Ω^{-1} or mho

OHM'S LAW

According to Ohm's law, the current flowing in a conductor is directly proportional to the potential difference applied across its ends provided the physical conditions and the temperature of conductor remain constant.



If $V_A = V_B \Rightarrow \Delta V = 0 \Rightarrow I = 0$

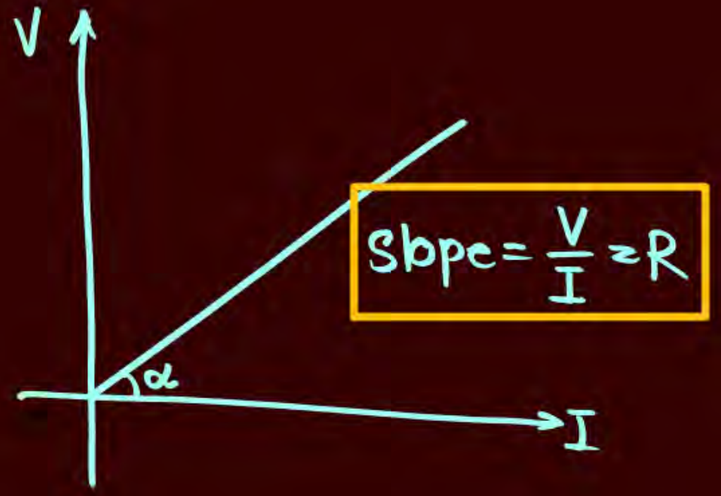
SSS Current flows from high potential to low potential

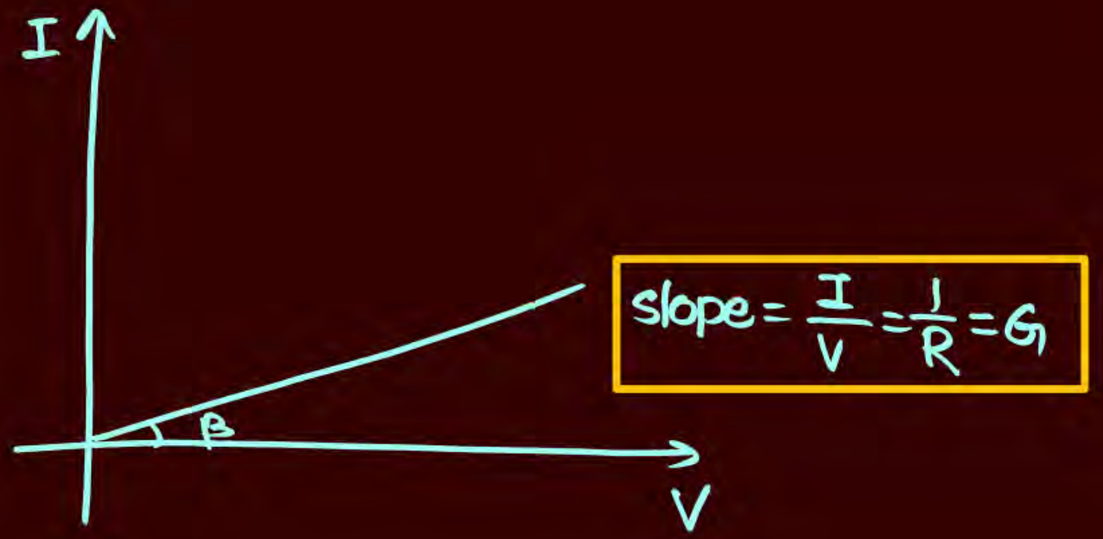
$\Delta V \propto I$

$\boxed{\Delta V = IR}$ or $\boxed{V = IR}$

R → Constant

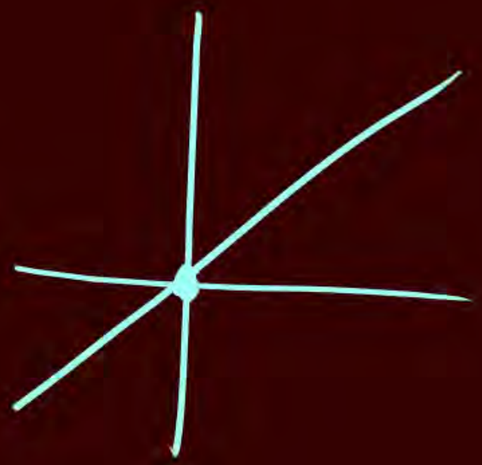
V → Potential difference.





$V = R\,I$

$y = mx$



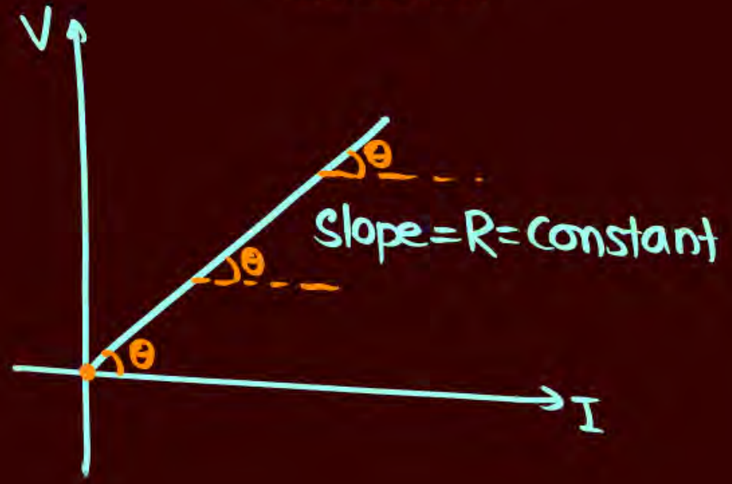
Ohmic Resistor

Follow Ohm's Law.

$V = IR$

↓

Constant

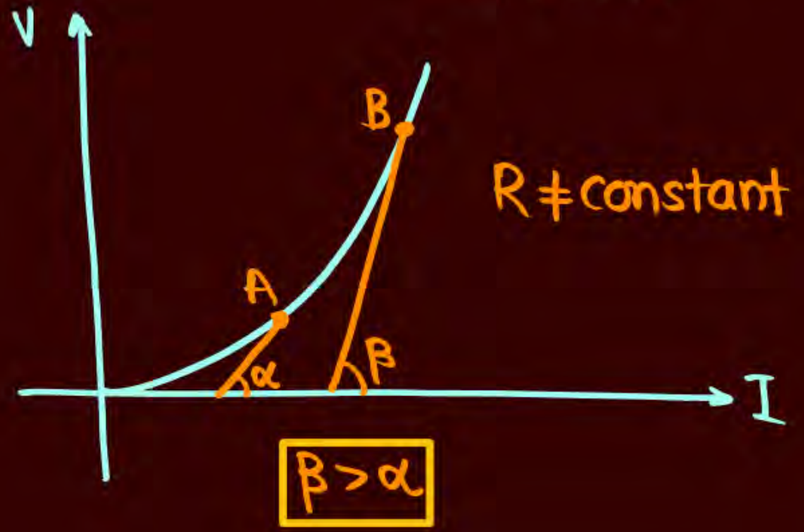


Non-Ohmic Resistor

→ Doesn't follow Ohm's law.

$V = IR$

↳ variable.



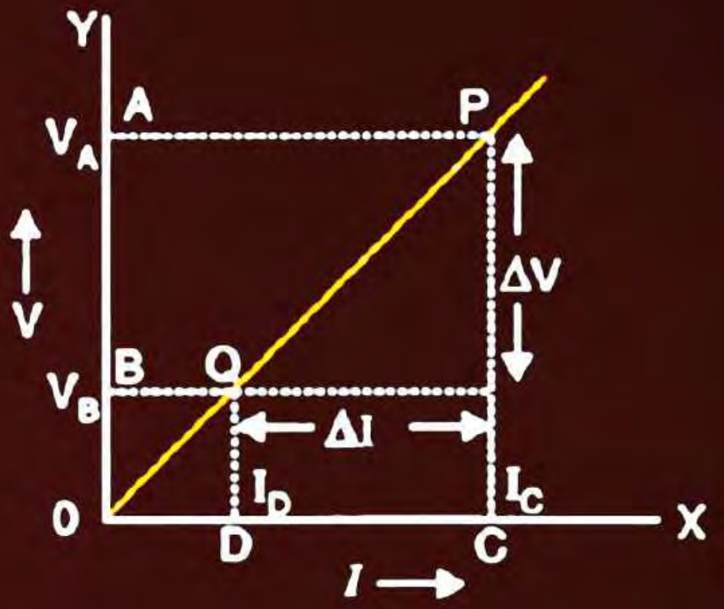
$\beta > \alpha$

OHMIC RESISTORS

The **conductors which obey the Ohm's law** are called the ohmic resistors (or linear resistances).

Examples: all metallic conductors (such as silver, aluminium, copper, iron, etc.), nichrome, copper sulphate solution with copper electrodes, and dil. sulphuric acid, etc. at a constant temperature.

For such resistors, a graph plotted for the potential difference V against current I is a straight line passing through the origin and the resistance R is same irrespective of the value of V or I (i.e., the ratio V/I is constant for all values of V or I).

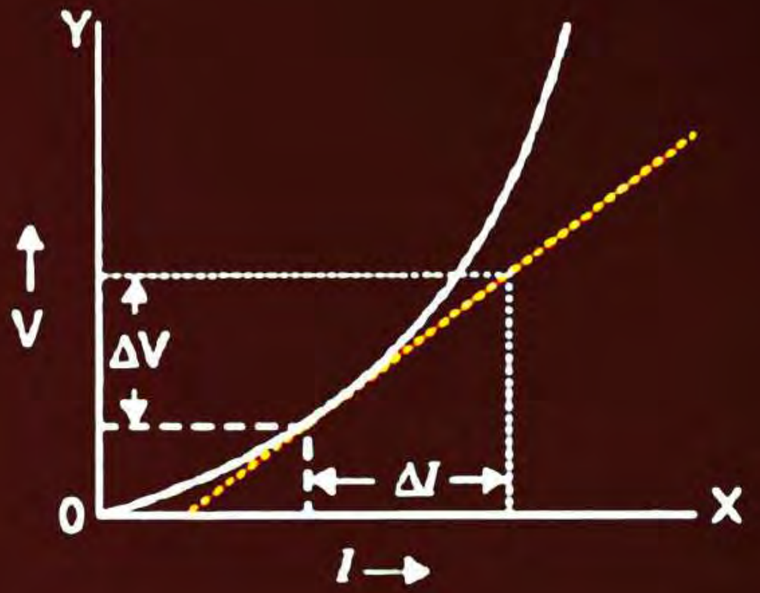


NON-OHMIC RESISTORS

The conductors which **do not obey the Ohm's law** are called the nonohmic resistors (or non-linear resistances).

Examples: LED, solar cell, junction diode, transistor, filament of a bulb, etc.

For these conductors, the graph plotted for the potential difference V against current I is not a straight line, but it is a curve. Fig. shows a $V-I$ graph in case of a junction diode.



Distinction between the ohmic and non-ohmic resistors

HW

Ohmic resistor	Non-ohmic resistor
1. It obeys the Ohm's law. i.e., V/I is constant for all values of V or I.	1. It does not obey the Ohm's law i.e., V/I is not same for all values V or I.
2. The graph for potential difference V versus current I is a straight line.	2. The graph for potential difference V versus current I is not a straight line.
3. The slope of $V - I$ graph is same at all values of V or I at a given temperature	3. The slope of V-I graph is different at different values of V or I even at a given temperature.
Examples : All metallic conductors such as silver, iron, copper, nichrome, electrolyte with suitable electrodes, etc.	**Examples :** Junction diode, LED, transistor, filament of a bulb, etc.



HW

CHOICE OF MATERIAL OF A WIRE

(1) **The wires used for electrical connections and for power transmission:** Such wires should possess negligible resistance. Due to low (or negligible) resistance of connection wires, the current in circuit remains unaffected, and the loss of energy due to heating is prevented. Hence they are made of materials such as **copper or aluminium**, whose specific resistance is very small. Further they are **made thick** so that their resistance becomes low.

Note: Though specific resistance of **silver** is less than that of copper, but because of its high cost, silver is not used.

(2) **The resistance wires (or standard resistors):** Such wires are of high resistance and their resistance almost does not change due to change in temperature. They are made from **alloys such as manganin, constantan,** etc. for which the specific resistance is high and the effect of change in temperature on their resistance is negligible.



CHOICE OF MATERIAL OF A WIRE

(3) **A fuse wire:** It is made from an **alloy of lead and tin** because its melting point is low and its specific resistance is more than that of copper or aluminium so that the resistance of a short and thin fuse wire is high to the extent that it permits current up to its safe limit to pass through it. An excessive current melts it so that it blows off.

(4) **The filament of an electric bulb:** A **tungsten wire** is used because it has a high melting point.

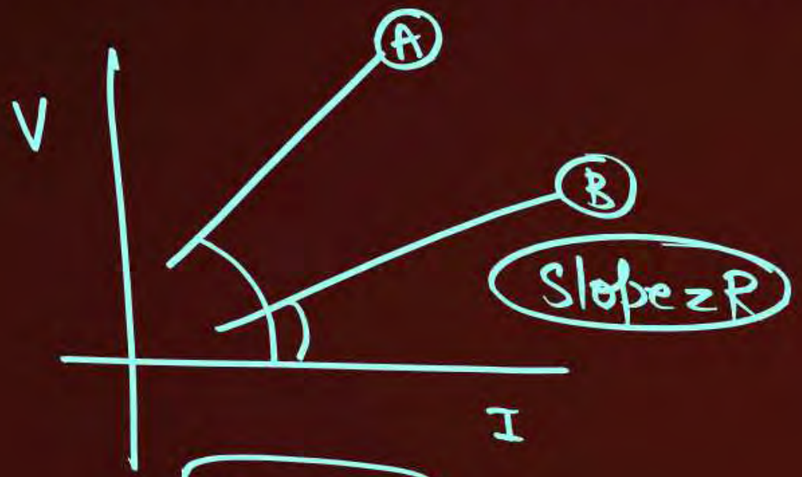
(5) **The heating element in appliances such as heater, toaster, oven, etc.:** A **nichrome wire** is used because the specific resistance of nichrome is high and its resistance increases to a great extent with the increase in temperature.

QUESTIONS

Fig. shown $I - V$ graph for two conductors A and B.

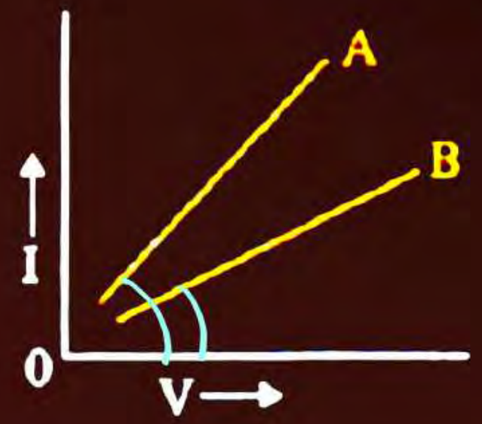
(i) Which of the conductor is ohmic? Both

(ii) Which conductor has more resistance? Give reason to your answer.



$m_A > m_B$

$R_A > R_B$



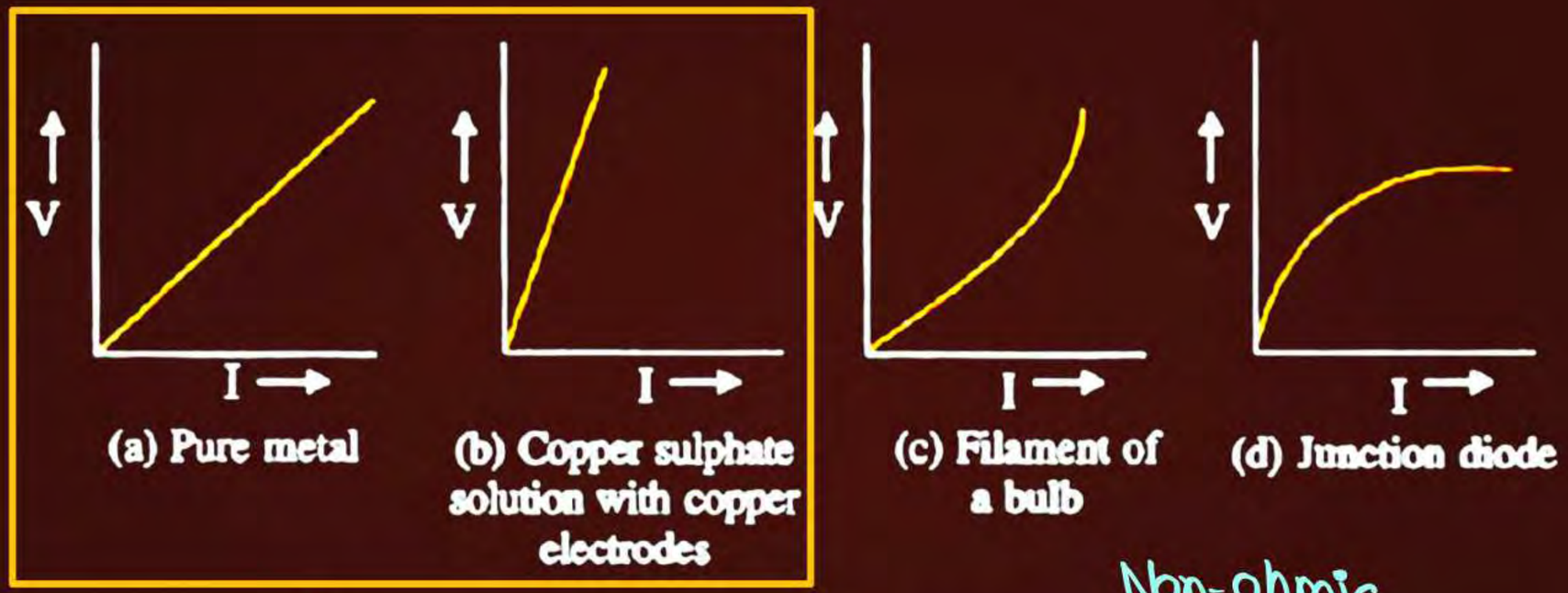
$$\text{Slope} = \frac{I}{V} = \frac{1}{R}$$

$m_A > m_B$

$R_A < R_B$

QUESTIONS

Fig. shows V-I graphs experimentally obtained in different cases. Select the graphs for ohmic and non-ohmic resistors.



Ohmic

Non-ohmic



QUESTIONS

Calculate the potential difference required across a conductor of resistance 5Ω to pass a current of 1.5 A through it .

$V = IR$

$= 1.5 \times 5$

$= 7.5\ V.$



QUESTIONS

An electric gadget draws a current 200 mA from a battery of 12 V. Find its resistance.

$V = IR$

$12 = 200 \times 10^{-3} \; R$

$R = \frac{12}{0.2} \; \Omega.$

QUESTIONS

Imp

A torch bulb when cold has a resistance 1Ω. It draws a current 300 mA, when glowing from a source of 3 V. Calculate the resistance of the bulb when glowing and explain the reason for the difference in resistance.

$R_{cold} = 1\Omega$

$T \uparrow R \uparrow$

$I = 300 \times 10^{-3}$ A.

$V = 3$ volts.

$V = IR$

$3 = 300 \times 10^{-3} R$

$R = 10\Omega$

QUESTIONS

Calculate the resistance of 1 km long copper wire of radius 1 mm. (Specific resistance of copper is $1.72 \times 10^{-8}\Omega m$).

$l = 1000\,m.$

$r = 1 \times 10^{-3}\,m.$

$\rho = 1.72 \times 10^{-8}\,\Omega m.$

$$R = \rho \frac{l}{A}$$

$$= 1.72 \times 10^{-8} \times \frac{1000}{\pi (10^{-3})^2}$$

$$= 1.72 \times 10^{-8} \times \frac{1000}{\pi \times 10^{-6}}$$

$$= 0.55 \times 10$$

$$= 5.5\,\Omega.$$

QUESTIONS

When a potential difference of 2 volt is applied across the ends of a wire of 5 m length, a current of 1 A flows through it. Calculate:

(i) the resistance per unit length of the wire,

(ii) the resistance of 2 m length of the wire,

(iii) the resistance across the ends of the wire if it is doubled on itself.

$V = IR$

$2 = 1R$

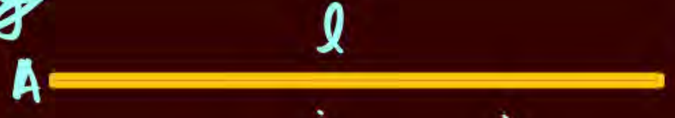
$R = 2\Omega$

$5\,m \longrightarrow 2\Omega$

$1\,m \longrightarrow \frac{2}{5}\Omega$

$2\,m \longrightarrow 2 \times \frac{2}{5} = \frac{4}{5}\Omega$

Folding



$$R = \rho \frac{l}{A}$$



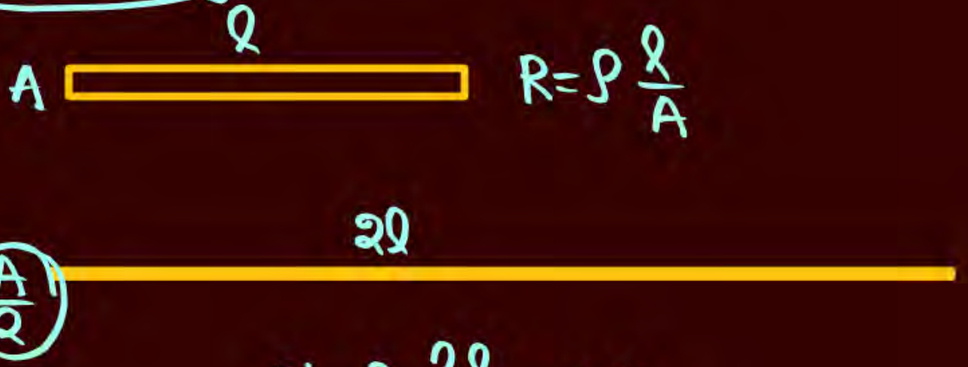
$$R' = \rho \frac{l/2}{2A}$$

$$R' = \frac{1}{4} \rho \frac{l}{A}$$

$$\boxed{R' = \frac{R}{4}}$$



Stretching



$$R' = \rho \frac{2l}{\frac{A}{2}} = 4R$$



QUESTIONS

Which of the following is an ohmic resistance?

1. LED
2. junction diode
3. filament of a bulb
4. nichrome.

QUESTIONS

The graph plotted for potential difference:(V) against current (I) for ohmic resistors is:

[ICSE Sem-2, 2022]

1. a curve passing through the origin
2. a straight line not passing through origin,
3. a straight line passing through origin
4. a circle centered at the origin.



QUESTIONS

1 C charge means a deficit of _____ electrons.

1. 1
2. 10^{19}
3. 6.25×10^{18}
4. 1.6×10^{-19}

QUESTIONS

Unit of specific resistance is:

1. Ω
2. Ωm
3. Ωm^{-1}
4. $\Omega^{-1} m$

$$R = \rho \frac{l}{A}$$

$$\rho = \frac{RA}{l}$$

$$= \Omega \frac{m^2}{m}$$

$$= \Omega m$$



QUESTIONS

A substance of zero resistance at a very low temperature is:

1. conductor
2. resistor
3. electrolyte
4. super conductor

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.

Thank You